

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	6aba25588343c5cff063937b
Project Name	Comic Craft AI
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Prettier (JS) and Black (Python) enforce indentation.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate frontend, backend and ai-service folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names like storyPrompt, panelImageUrl.
4	Function / Method Names	Functions are descriptively named	Yes	e.g. generateComicScript(), saveComic().
5	Code Comments	Inline and block comments explain logic	Partial	Complex logic commented; some helpers pending.
6	Modular Design	Code is split into reusable functions/modules	Yes	Auth, generation and export are separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Partial	Duplicates removed; few unused imports left.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	try/catch and error middleware for API and AI calls.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Auth Module (register, login, JWT)	Node.js, Express, JWT	Registration, Login, Admin Dashboard	High

2	Comic Script Generator	Python, LLM API	Story input, Panel editor	Medium
3	Panel Image Generator	Python, AI image model	Comic generation, Panel regenerate	Medium
4	UI Component Library (buttons, cards, modals)	React, Tailwind CSS	All web app pages	High
5	Export Service (PDF / image)	Node.js	My Library, Share link	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear folder structure and consistent formatting.
Readability	4	Descriptive names; code is easy to follow.
Reusability	4	Auth, UI and generation logic reused across features.
Documentation / Comments	3	Comments present; more docstrings to be added.
Overall Score	3.8 / 5	Good quality; documentation is the main improvement area.